

Lab 18 - Views, Partial & Tag Helpers

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Assignment: Lab 18 | ITI Summer Training - Web Development Using .NET

Fingerprinted values for Lab ID 31

Variable Name	Formula / Derivation	Result Value
CHIP_YEAR	(Lab ID mod 4) + 1 = 3 + 1	4
CHIP_LABEL	Final (since CHIP_YEAR is 4)	Final

Part A - Verify and Orient

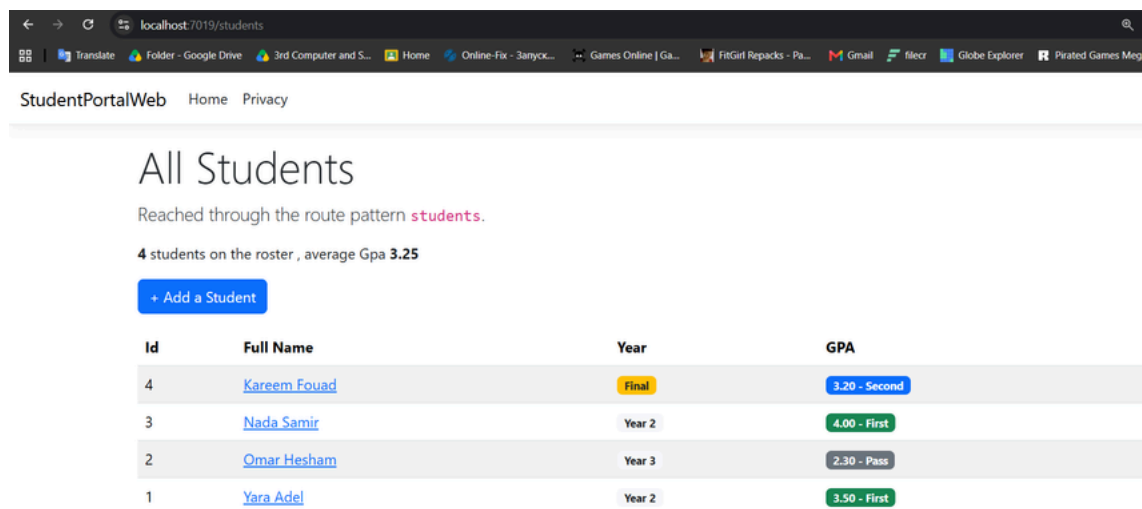
Verified that basic view tasks function correctly. Added Razor comment holding personal values to `_StudentRow.cshtml`.

- **Why does Model mean one Student in `_StudentRow` and a whole list in `Index`?** The model type is declared via the `@model` directive. `Index.cshtml` has `@model List<Student>` configuring its model as a collection, while `_StudentRow.cshtml` has `@model Student` configuring its model as a single Student object.

Part B - Say it once, everywhere

DeduplicatedByYear.cshtml, Honours.cshtml, and Searching.cshtml to call `_StudentRow` partial and deleted the duplicate `<tr>` row structures.

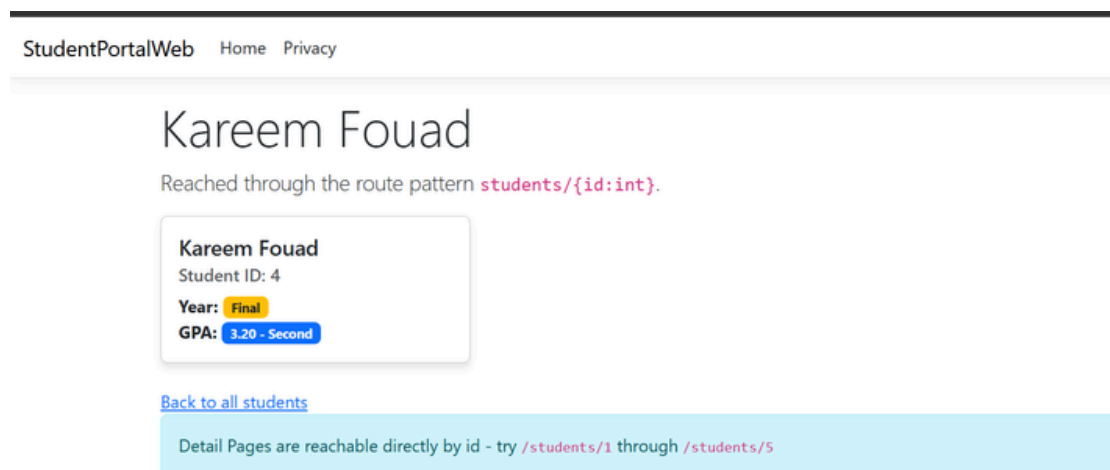
- **What would happen if you called the partial with no model attribute?** The partial would inherit the parent's model (`List<Student>`). Since `_StudentRow.cshtml` expects a single Student, this results in a runtime `InvalidOperationException` due to type mismatch.



Part C - A second partial, a different shape

Created `_StudentCard.cshtml` partial. Modified `Details.cshtml` to render it instead of the old table layout. The card successfully renders the GPA using the `gpa-badge` tag helper.

- **Could one page use both partials at once?** Yes. Both accept a `Student` model but generate different HTML layout representations (a table row vs. a card), allowing us to use them concurrently on the same page.

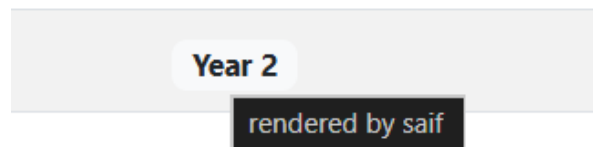


Part D - Your own tag helper

Created `YearChipTagHelper` mapping 'year-chip' element. It renders a yellow 'Final' badge for year 4 and grey 'Year n' badges for others. Configured title attribute as 'rendered by saif'.

Id	Full Name	Year
4	Kareem Fouad	Final
3	Nada Samir	Year 2
2	Omar Hesham	Year 3
1	Yara Adel	Year 2

Year



Part E - Prove it, and break it

HTML source: `<span class="badgebg-warning text-dark" title="rendered by saif">Final</span>`

- Broken symptoms:

- (a) Removing `addTagHelper` renders custom raw tags `<year-chip for='4'>`;
- (b) Removing `TagMode` renders self-closing empty spans `<span ... />`;
- (c) Removing hyphen renders raw unrecognized tag `<yearchip for='4'>`.

- **Common theme:** All three fail silently without compile or run-time exceptions, outputting invalid tags to the browser.

Part F - Wrap-Up Reflection

1. **DerivedValues:** LabID:31 | CHIP_YEAR: 4 | CHIP_LABEL: Final | First Name: saif.

2. Duplication Parallel:

- In Session 15, the database connection string and options were duplicated inside context and Program.cs. This was solved by configuring options inside Program.cs and injecting DbContextOptions into the context constructor via Dependency Injection.

3. Shared Extensibility Pattern:

- All three require us to inherit from a base class or implement an interface, override a single core method (Match, IsValid, or Process), and register the class, allowing the engine to execute it dynamically during the request lifecycle.

4. Rule vs. Label Duplication:

- gpa-badge contains a rule (GPA threshold ranges for grades), while year-chip contains a label (display name mapping for year numbers). Duplicating the rule in view files is far more dangerous because changes to grading standards would require searching and modifying all views, risking silent drifts. Presentation concern vs. business concern.